Improving Factuality of Large Language Models through Retrieval-Augmented Generation and Self-Verification

NLP with LLMs — Course Project Proposal

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Project Objectives

- Analyze the trade-off between factuality improvement and computational cost.
- Perform ablation experiments to quantify the contribution of retrieval, self-consistency, verification, and correction.
- Develop an evidence-based correction module for unsupported or contradicted claims.
- Develop a claim-level verification module using retrieved evidence and NLI/LLM-based verification.
- Compare direct prompting, Retrieval-Augmented Generation (RAG), and self-consistency-based approaches for reducing hallucinations.
- Evaluate the factuality and hallucination rate of open-source LLMs on established NLP benchmarks.

1) Motivation

Large Language Models (LLMs) can generate fluent and convincing responses that contain unsupported, fabricated, or contradictory information, commonly referred to as hallucinations. This limits their reliability in applications such as question answering, education, research assistance, and information retrieval. Existing approaches such as Retrieval-Augmented Generation (RAG) reduce factual errors by supplying external evidence, but retrieval alone does not guarantee that the generated response is fully supported by the retrieved information. Recent work such as SelfCheckGPT and Chain-of-Verification shows that consistency checking and explicit verification can further improve factuality.

This project proposes a practical and experimentally rigorous pipeline that first generates an answer, detects unsupported claims, retrieves or verifies relevant evidence, and then revises the answer. The central research question is: how much can hallucination detection and self-verification improve factuality compared with a standard LLM and a conventional RAG system, and what is the associated cost in latency and computation?

Rather than developing another standalone RAG application, this project will conduct a controlled comparison of different LLM-based strategies and use ablation experiments to identify which components contribute most to factuality improvement.

2) Problem Description

Given a user question, an LLM may produce an answer containing a mixture of correct and hallucinated statements. The goal is to build an NLP system that:

1. Generates an initial answer to a question.
2. Identifies claims in the answer that may be unsupported or inconsistent.
3. Retrieves relevant evidence or performs independent verification.
4. Assigns a hallucination/factuality decision to each claim.

5. Produces a corrected final answer with evidence/citations where possible.

The project will compare several systems: (i) a standard prompting baseline, (ii) RAG, (iii) self-consistency-based hallucination detection inspired by SelfCheckGPT, (iv) Chain-of-Verification-style reasoning, and (v) a proposed hybrid detector + evidence retrieval + correction pipeline.

The main hypothesis is that combining retrieval with independent verification and claim-level correction will reduce hallucination rate more reliably than using generation or retrieval alone, while maintaining acceptable answer quality and computational cost.

3) Dataset to be Used

Primary datasets:

1. HaluEval
HaluEval is a large hallucination evaluation benchmark containing 35K examples: 5,000 general user queries and 30,000 task-specific examples spanning question answering, knowledge-grounded dialogue, and summarization. It contains hallucinated and non-hallucinated responses and is directly suited for training/evaluating hallucination recognition.

2. RAGTruth
RAGTruth is a corpus of nearly 18,000 naturally generated RAG responses with human annotations at case and word level. It is especially valuable because hallucinations can still occur even when a retrieval component is present. It can be used to evaluate the proposed detector and the RAG-based correction stage.

3. FEVER
FEVER contains 185,445 claims classified as Supported, Refuted, or NotEnoughInfo, together with evidence for supported/refuted claims. It provides a clean fact-verification benchmark for evaluating whether the verification module can correctly identify factual status.

4. TruthfulQA
TruthfulQA contains 817 questions designed to test whether models reproduce common misconceptions. It will be used as an additional out-of-domain test set for truthfulness.

Recommended experimental split:
- Train/validation: HaluEval and/or RAGTruth training portions.
- Development: held-out HaluEval/RAGTruth samples.
- Evaluation: RAGTruth test set + FEVER + TruthfulQA.
- A small manually checked subset may be created to test end-to-end correction quality and citation correctness.

4) Methodology

The proposed system will be implemented as a modular pipeline:

Stage 1 — Initial Generation
A question is supplied to an instruction-tuned LLM, which generates an answer without external verification.

This serves as the baseline.

Stage 2 — Claim Extraction
The answer is segmented into atomic or near-atomic factual claims. Each claim is treated as a unit for verification.

Stage 3 — Hallucination Detection
Three complementary strategies will be investigated:
• Self-consistency: generate multiple responses and measure semantic/NLI agreement, inspired by SelfCheckGPT.
• Evidence-based verification: retrieve passages relevant to each claim and determine whether the evidence supports, refutes, or does not establish the claim.
• LLM-based verification: prompt a verifier model to assess the claim against the retrieved evidence.

Stage 4 — Retrieval
A dense retriever such as a Sentence-BERT/contriever-style encoder will retrieve top-k evidence passages from a suitable corpus. For FEVER-style experiments, the provided Wikipedia evidence can be used; for RAGTruth, the provided retrieved context can be used.

Stage 5 — Correction
Claims classified as unsupported or contradicted will be removed, rewritten, or replaced using evidence. A Chain-of-Verification-style procedure will be tested as one correction strategy.

Stage 6 — Final Answer
The corrected claims are composed into a final answer. Where evidence is available, citations are attached to the relevant claims.

Stage 7 — Evaluation
The systems will be compared on:
• Hallucination detection: accuracy, precision, recall, F1, AUROC/AUPRC where applicable.
• Factuality: claim-level support rate / FACTSCORE-style atomic factual precision.
• End-to-end QA: exact match/F1 where applicable and human/LLM-assisted factuality evaluation.
• Citation quality: evidence relevance and citation correctness.
• Efficiency: latency, number of model calls, token usage, and approximate inference cost.

Key ablations:
A. LLM only
B. LLM + RAG
C. LLM + self-consistency
D. LLM + verification
E. LLM + RAG + verification
F. LLM + RAG + verification + correction

This structure allows the project to answer not just whether the final system works, but which component actually reduces hallucinations.

How This Work Helps Now and in the Future

Immediate learning and usefulness:
The project will provide hands-on experience with modern NLP/LLM pipelines rather than only API-based chatbot development. It will cover retrieval, embeddings, prompting, NLI-style verification, evaluation, dataset handling, model inference, and error analysis. It also develops the ability to conduct controlled experiments and interpret research results.

Future usefulness:
Reliable LLMs are increasingly important in enterprise search, coding assistants, research tools, education, and decision-support systems. Skills in factuality evaluation, RAG, verification, and model reliability are therefore directly transferable to LLM engineering and research roles. The project also creates a strong foundation for future work in trustworthy AI, LLM evaluation, RAG systems, and efficient model deployment.

5) References

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Expected Deliverables

- A working hallucination detection and correction pipeline.
- A reproducible experimental comparison of at least 4 system variants.
- Quantitative tables/plots for detection and factuality performance.

- Ablation study and qualitative error analysis.
- An interactive demonstration showing initial vs verified/corrected answers.